Disease	Any condition that upsets the harmonious functioning of the organism through physical damage or impairment of a chemical process (i.e. the disturbance of homeostasis).
Pathogen	Biological agent that causes disease
Signs	Features that may be observed in the patient by others (e.g. cough, weight loss).
Symptoms	Features that are apparent only to the patient (e.g. weakness, tiredness, headache).
Parasite	Organism that lives on or in a host and gets its food from (or at the expense of) its host
Germ Theory of Disease	Microorganisms are the cause of many diseases
Spontaneous generation	Theory that living organisms can spontaneously arise from non-living matter
Reservoir	Place in which the pathogen normally lives and reproduces
Carrier	An individual with a disease who does not exhibit signs of the disease but can still infect others
Incubation period	Time between entry of pathogen and the onset of symptoms of the disease
Zoonotic disease/zoonoses	Infectious diseases that have jumped from a non-human animal to humans
Direct transmission	The spread of disease from one host to another, without an intermediate organism.
Direct contact	Skin-to-skin contact made with an infected individual
Vector transmission	Involves the spread of disease from host to host by means of a vector
Antigen	A molecule that is not recognised by the immune systems as 'self'
First line of defence	Series of barriers that prevent the entry of pathogens
Granuloma	
Cytokines	Signalling molecules that enhance the immune response
Innate immunity	The first and second lines of defence; these are the defences that are present at birth
Adaptive immunity	The third line of defence. It is created in response to an antigen, either through natural exposure or through vaccination
Antibodies	Globular proteins that bind to a particular antigen and mark it for elimination
Clonal Selection Theory	In a pre-existing group of B cells, a specific antigen only activates its counteracting B cell, causing that particular cell to multiply for antibody production
Vaccination	The process where an individual is treated with a vaccine in order to produce immunity against a disease
Artificial active immunity	Using a vaccine to stimulate the body's own immune response
Artificial passive immunity	Involves an injection of antibodies in the form of an 'anti-serum'
Herd immunity	

Quarantine	
Public health campaigns	Activities carried out by the Australian Government, state, territory and local government agencies, non-government agencies and private health professionals, aiming to improve public health and quality of life; by focusing on prevention, promotion and protection through the factors and behaviours that cause illness and injury
Pesticides	Chemicals that destroy organisms considered to be pests
Antivirals	Drugs that are used to cure or control viral infections
Antibiotics	Chemicals that inhibit, destroy or prevent the growth of bacteria
Bactericidal antibiotics	kill microbes directly by damaging the cell wall (eg. penicillin) or cell membrane (eg. nystatin)
Bacteriostatic antibiotics	Do not kill the bacteria directly, but prevent them from replicating (eg. rifampin) or inhibit translation (eg. streptomycin).
Epidemic	Widespread occurrence of an infectious disease affecting many people at the one time
Pandemic	Widespread occurrence in many parts of the world of an infectious disease affecting many people at the one time
Endemic	Widespread occurrence of an infectious disease affecting many people at the one time, that is always present in that part of the world
Sporadic	Occurring in different places at different times
Treatment	Either cures the disease or relieves its symptoms, once an organism has the disease
Control	Involves reducing its spread through the population of organisms, once it is already present
Prevention	Involves the use of strategies that stop the occurrence of disease in organisms
Management	Most commonly involves programs that improve the outcomes of chronic conditions and improve the quality of life of sufferers
Bush medicine	The use of herbs and plants to treat illnesses, both physical and mental
DNA helicase	Unwinds the two DNA strands of the double helix by breaking bonds between bases.
RNA primase	Adds a short, complimentary RNA primer to exposed 3’ end of each strand.
DNA polymerase III	Attaches free nucleotides to the primers on each exposed strand according to the base pair rule. 5’ to 3’ direction.
DNA polymerase I	~~Removes primers from the fragments and~~ inserts correct DNA nucleotides.
Exonuclease	Removes RNA primers.
DNA ligase	Links the fragments of the lagging strand together to form a continuous DNA strand.
Binding proteins	Attach to separated DNA strands to stop them reattaching
Gene	A sequence of bases along a DNA strand that provides the coded information required to make a specific polypeptide for the cell
Polypeptide synthesis	Process of producing a polypeptide using the information in a gene
Transcription	The process by which the information on the DNA is copied onto an mRNA molecule

Translation	The process by which the mRNA molecule is used to make a new polypeptide chain
Exons	Protein-coding sections of DNA code for the production of a polypeptide.
Introns	Non-coding sections of DNA whose biological functions remain unknown.
Alternative splicing	
Phenotype	Observable characteristics of an organism
Genotype	Set of genes the organism carries
Epigenetics	The study of how environmental factors can change the way genes are expressed
Cline	Continuous, or near continuous, gradation of a phenotypic character within a species, associated with an environmental variable
Fertilisation	Union of sperm and ovum
Codominance	Both alleles expressed in phenotype
Incomplete dominance	Phenotype is a blending of alleles
Multiple alleles	More than two alleles for a gene
Single nucleotide polymorphism (SNP)	Substitution of a single nucleotide
HapMap	Catalogue of common SNPs
DNA sequencing	The process of determining the precise order of bases within a DNA molecule
Genome	An organism's complete set of DNA, including all of its genes
Genomics	The study of individual genes and their functions, including inheritance
DNA profiling/ fingerprinting/testing	A technique that produces a unique genetic profile or 'fingerprint' for each individual
Reproduction	A organism producing more of its species to replace those that die
Pollination	The transfer of pollen from the anther onto the stigma (usually of a different flower) of the same species
Spermatogenesis	Production of sperm
Oogenesis	Production of oocytes
Gametogenesis	Production of gametes
Menstruation	
Mutation	Permanent change to organism's DNA
Mutagen	Chemical, radiation or other naturally occurring condition that is capable of causing a mutation
Mutagenesis	Process of causing a mutation
Transposon	Sections of DNA that can multiply or relocate themselves within the genome
Retrovirus	Virus whose genetic information is in the form of RNA

Point mutation	Changes in the DNA sequence of a single gene
Base substitution	Single nucleotide substituted with another
Missense mutation	New codon results in the inclusion of an amino acid that is chemically different to the original
Neutral mutation	New codon results in the inclusion of an amino acid that is chemically similar to the original
Silent mutation	New codon results in the same amino acid as the original
Frameshift mutation	
Live vaccine	Use a weakened (attenuated) form of the pathogen, that has had its disease-causing capacity reduced by heating, drying, chemical treatment or irradiation, eg. measles
Inactivated vaccine	Consist of dead pathogens, eg. polio
Subunit vaccine	Consist of the pathogen's antigens, rather than the pathogen itself, eg. human papillomavirus
Toxoid vaccine	Use a toxin produced by the pathogen, eg. diphtheria and tetanus
mRNA vaccine	Consist of viral RNA enclosed in a lipid nanoparticle or carrier virus, eg. Pfizer and Moderna COVID-19
Frameshift mutation	One or more nucleotides are deleted or inserted
Nonsense mutation	Early production of a stop codon leads to a truncated polypeptide
Chromosomal mutations/aberrations	Mutations that affect large regions or whole chromosomes
Rearrangement	Segments of chromosome are rearranged during crossing over in meiosis
Deletion	2 breaks in chromosome and middle bit lost, ends rejoin, or 1 break and then end snaps off.
Inversion	Segment of the chromosome removed, rotated 180 and replaced. Gene may be cut in half or removed from promoter region
Translocation	Movement of segment to a non-homologous chromosome
Duplication	Segment lost from one chromosome and added to its homologue
Aneuploidy	Chromosome number is not an exact multiple of haploid number.
Syndrome	
Polyploidy	Organism has one or more extra sets of all chromosomes
Colchines	Chemicals that disrupt the formation of the mitotic spindle; used to induce polyploidy
Germline cells	Cells that produce gametes
Proto-oncogene	Gene that controls the rate of cell division
Coding DNA segments/ sequence	Portion of a gene's DNA that codes for a polypeptide
Non-coding DNA	Does not code for a polypeptide; regulates genes, switches genes on/off, codes for rRNA and small nuclear RNA (determines which introns are spliced)

Junk DNA	No known function
Transposition	The jumping process of a transposon relocating within the genome
Gene pool	Total collection of genes in a population at any one time
Gene flow	Alleles from one population are introduced to another by migration
Genetic drift	Chance events cause unpredictable changes in a population's allele frequency from one generation to the next
Bottleneck effect	Large population suddenly and drastically reduced in size
Founder effect	New population established by a very small number of individuals from a larger population
Biotechnology	Any technological application that uses plants, animals or microorganisms to make or modify products or processes that benefit humans
Technology	
Reproductive technology	Technologies that artificially intervene with the natural processes of reproduction
Whole organism cloning	Process of producing an organism or organisms that are genetically identical to the original
Gene cloning	Use of genetic engineering to produce many copies of identical genes
Transgenic organisms	Organisms that, through genetic engineering, have had [genetic material] from another species inserted into their genome
Biodiversity	Variety of life on Earth
Genetic diversity	
Species diversity	
Ecosystem diversity	
Genetic engineering	
Homeostasis	Process by which organisms maintain a relatively stable internal environment
Ectothermic organism	Organism whose metabolism generates little heat
Endothermic organism	Organism whose body temperature remains relatively constant
Adaptation	Inherited characteristic, process or behaviour that is of benefit to the organism, and aids its survival in a particular habitat
Behavioural adaptation	The activities and behaviours of the organisms, eg. huddling
Structural adaptation	The physical characteristics of an organism, eg. body size
Physiological adaptation	The process carried out by the structures, eg. sweating
Conduction	Direct heat transfer

Radiation	Indirect heat transfer
Sensory neuron	Arise from sensory receptors and carry messages to the CNS for processing, keeping the CNS aware of internal and external environments
Interneurons	Located in the CNS, carry impulses from sensory to motor neurons
Motor neurons	Carry impulses from the CNS to effectors.
Reflex	Automatic response to a stimulus involving a small number of neurons
Synapse	Space between the axon of one neuron and the dendrite or cell body of the next neuron
Hormone	Chemical messengers that act on specific target cells to cause a specific response for internal communication
Auxin	From roots and stems, stimulates the elongation of cells, resulting in root formation, shoot growth and bending towards the light
Cytokinins	Produced by roots, and transported to embryos and fruits, promote cell division, encouraging seed germination and seedling growth
Gibberellins	Produced by young roots and leaves, increase cell elongation and division, causing stem and leaf growth, and stimulating flowering a development
Abscisic acid	Produced by leaves, stems, roots and green fruit, inhibits germination. Ratio of [X] to gibberellins often determines whether a seed remains dormant or germinates
Ethylene	Produced by ageing fruits and leaves, triggers fruit ripening and other ageing processes. A changing ratio of auxin to [X], triggered mainly by shorter days, probably causes autumn colour changes and the loss of leaves from deciduous trees
Xerophyte	Plants adapted to low water conditions
Halophyte	Salt tolerant plant
Non-infectious disease	Disease that is not caused by pathogens and is not transmittable from one individual to another (this includes hereditary diseases)
Cancer	Cells which multiply uncontrollably due to damage in the DNA.
Malignant	Cancers that spread into surrounding areas, or different parts of the body
Benign	Cancers that do not spread beyond the immediate area
Malnutrition	Deficiencies, excesses, or imbalances in a person’s intake of energy and and/or nutrients
Descriptive studies	Study distribution in relation to age, ethnicity, sex etc.
Intervention studies	Evaluate treatment and prevention methods (eg. clinical trials)
Analytical studies	Determine the cause, risk and protective factors
Observational (case-controlled) studies	A type of analytical study where diseased individuals (cases) are compared to healthy individuals (controls). Retrospective, ie look back at the past exposure and medical history of an individual ro draw conclusions relating to the cause and risk factors of a disease
Experimental (cohort) studies	A type of analytical study where individuals who are and aren’t exposed to a particular risk factor. Prospective, ie look forward over a period time to see who develops the disease to draw conclusions about the cause and risk factors of a disease

Educational programs/campaigns	Carried out by Australian Government, state, territory and local government agencies, non-government agencies, and private health professionals. They aim to raise awareness of the risk factors associated with particular diseases, to get people to change their behaviour.
Biofortification	Increases the nutritional value of crops through selective breeding or genetic modification
Astigmatism	Cornea more curved in one direction that another
Cataract	The eye’s crystalline lens becomes cloudy or yellow, preventing light rays from properly focusing inside the eye
Crossing over	The exchange of chromosomal segments between non-sister chromatids of a homologous chromosome pair during prophase I of meiosis
Random segregation	The process by which homologous chromosome pairs are separated and randomly segregated into haploid cells during meiosis.
Independent assortment	
Infectious disease	
Evolution	The gradual change in a population over a long period of time by natural selection.
Natural selection	The process through which populations of living organisms adapt and change through environmental/selective pressures.